

ITSimplera Institute



Submitted By: Muhammad Yasir Khan

No: NETB01-4817

Domain: Network Administration

Date of Submission: 8th August, 2026

Supervisor: Jawad Qayum (Senior Network Administrator)

Company: IT-SIMPLERA Institute

Department: Network Administration

Contents

Week 6: Enterprise WAN Technologies and High Availability	3
1. Assignment Overview	3
2. Topology Details	3
3. IP Addressing Table	3
4. Protocols Configured	4
PPP/CHAP:	4
GRE Tunnel:.....	5
OSPF:.....	5
Floating Static Routes:	5
NAT/PAT:.....	5
5. Verification Results	6
6. Troubleshooting	6

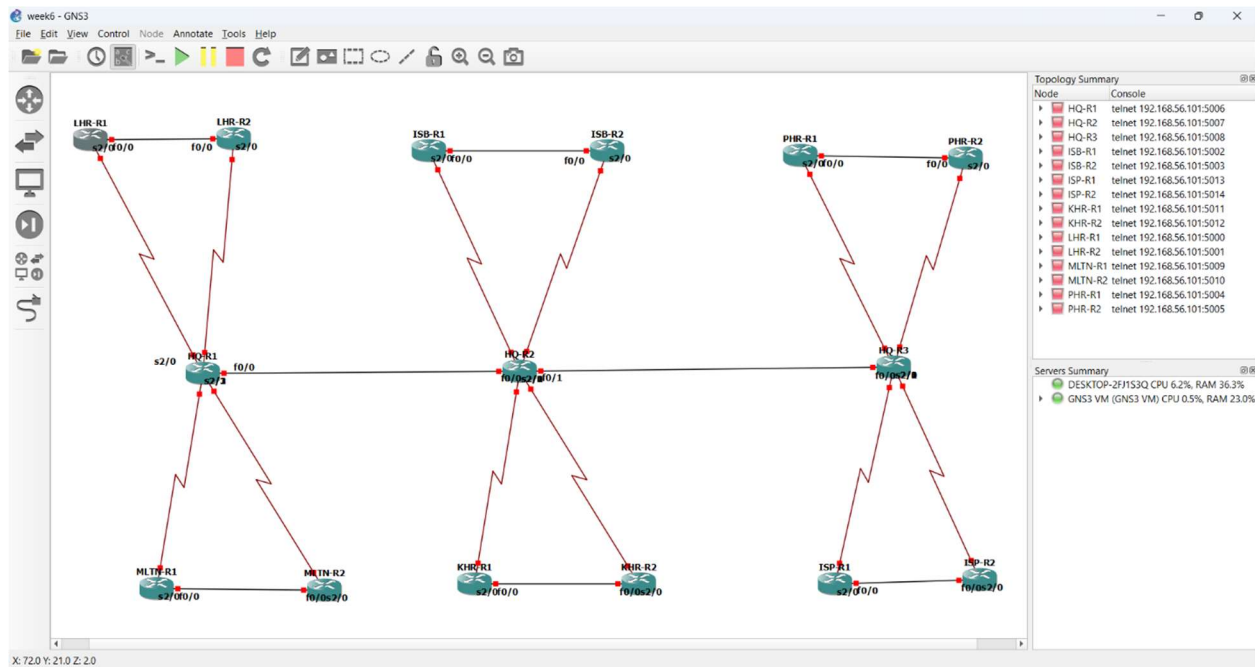
Week 6: Enterprise WAN Technologies and High Availability

1. Assignment Overview

Enterprise WAN infrastructure designed and implemented in GNS3 for ITSimplera Solutions connecting 6 branch offices through secure WAN links with high availability.

2. Topology Details

- **Total Routers: 15**
- **Branches:** Lahore, Islamabad, Peshawar, Karachi, Multan, ISP
- **HQ Routers:** HQ-R1, HQ-R2, HQ-R3
- **Design:** Hub-and-Spoke with redundant WAN links



3. IP Addressing Table

Device	Interface	IP Address	Purpose
HQ-R1	Loopback0	192.168.10.1/32	Router-ID
HQ-R1	Serial2/0	10.1.1.2/30	WAN to LHR-R1 Primary
HQ-R1	Serial2/1	10.1.2.2/30	WAN to LHR-R2 Backup
HQ-R1	Serial2/2	10.5.1.2/30	WAN to MLTN-R1 Primary
HQ-R1	Serial2/3	10.5.2.2/30	WAN to MLTN-R2 Backup
HQ-R1	Fa0/0	172.16.1.1/30	HQ Internal
HQ-R1	Tunnel0	172.16.12.1/30	GRE to LHR-R1
HQ-R2	Loopback0	192.168.10.2/32	Router-ID

HQ-R2	Serial2/0	10.2.1.2/30	WAN to ISB-R1 Primary
HQ-R2	Serial2/1	10.2.2.2/30	WAN to ISB-R2 Backup
HQ-R2	Serial2/2	10.4.1.2/30	WAN to KHR-R1 Primary
HQ-R2	Serial2/3	10.4.2.2/30	WAN to KHR-R2 Backup
HQ-R3	Loopback0	192.168.10.3/32	Router-ID
HQ-R3	Serial2/0	10.3.1.2/30	WAN to PHR-R1 Primary
HQ-R3	Serial2/1	10.3.2.2/30	WAN to PHR-R2 Backup
LHR-R1	Loopback0	192.168.20.1/32	Router-ID
LHR-R1	Serial2/0	10.1.1.1/30	WAN to HQ-R1
LHR-R1	Fa0/0	192.168.1.1/24	LAN
LHR-R1	Tunnel0	172.16.12.2/30	GRE to HQ-R1
LHR-R2	Loopback0	192.168.20.2/32	Router-ID
LHR-R2	Serial2/0	10.1.2.1/30	WAN to HQ-R1 Backup
ISB-R1	Loopback0	192.168.30.1/32	Router-ID
ISB-R1	Serial2/0	10.2.1.1/30	WAN to HQ-R2
ISB-R1	Fa0/0	192.168.2.1/24	LAN
ISB-R2	Loopback0	192.168.30.2/32	Router-ID
ISB-R2	Serial2/0	10.2.2.1/30	WAN Backup
PHR-R1	Loopback0	192.168.40.1/32	Router-ID
PHR-R1	Serial2/0	10.3.1.1/30	WAN to HQ-R3
PHR-R1	Fa0/0	192.168.3.1/24	LAN
PHR-R2	Loopback0	192.168.40.2/32	Router-ID
PHR-R2	Serial2/0	10.3.2.1/30	WAN Backup
KHR-R1	Loopback0	192.168.50.1/32	Router-ID
KHR-R1	Serial2/0	10.4.1.1/30	WAN to HQ-R2
KHR-R1	Fa0/0	192.168.4.1/24	LAN
KHR-R2	Loopback0	192.168.50.2/32	Router-ID
KHR-R2	Serial2/0	10.4.2.1/30	WAN Backup
MLTN-R1	Loopback0	192.168.60.1/32	Router-ID
MLTN-R1	Serial2/0	10.5.1.1/30	WAN to HQ-R1
MLTN-R1	Fa0/0	192.168.5.1/24	LAN
MLTN-R2	Loopback0	192.168.60.2/32	Router-ID
MLTN-R2	Serial2/0	10.5.2.1/30	WAN Backup
ISP-R1	Loopback0	192.168.70.1/32	Router-ID
ISP-R1	Serial2/0	10.6.1.1/30	WAN to HQ-R3
ISP-R2	Loopback0	192.168.70.2/32	Router-ID
ISP-R2	Serial2/0	10.6.2.1/30	WAN Backup

4. Protocols Configured

PPP/CHAP:

- Sab Serial links pe PPP encapsulation
- CHAP authentication with password: cisco123
- Clock rate 128000 on all HQ Serial interfaces

GRE Tunnel:

- Tunnel0 between HQ-R1 and LHR-R1
- Source: 10.1.1.2 → Destination: 10.1.1.1
- Tunnel IP: 172.16.12.1/30 ↔ 172.16.12.2/30

OSPF:

- Process ID: 1
- Area: 0
- All networks advertised
- Router-IDs: Loopback addresses

Floating Static Routes:

- Primary AD: 110
- Backup AD: 200
- Automatic failover on link failure

```
HQ-R1#
HQ-R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

192.168.30.0/32 is subnetted, 2 subnets
O   192.168.30.2 [110/75] via 172.16.1.2, 00:20:11, FastEthernet0/0
O   192.168.30.1 [110/75] via 172.16.1.2, 00:20:11, FastEthernet0/0
192.168.60.0/32 is subnetted, 2 subnets
O   192.168.60.1 [110/65] via 10.5.1.1, 00:20:41, Serial2/2
O   192.168.60.2 [110/65] via 10.5.2.1, 00:20:21, Serial2/3
192.168.10.0/32 is subnetted, 3 subnets
O   192.168.10.2 [110/11] via 172.16.1.2, 00:20:11, FastEthernet0/0
O   192.168.10.3 [110/21] via 172.16.1.2, 00:20:12, FastEthernet0/0
C   192.168.10.1 is directly connected, Loopback0
192.168.40.0/32 is subnetted, 2 subnets
O   192.168.40.1 [110/85] via 172.16.1.2, 00:20:13, FastEthernet0/0
O   192.168.40.2 [110/85] via 172.16.1.2, 00:20:13, FastEthernet0/0
172.16.0.0/30 is subnetted, 3 subnets
C   172.16.12.0 is directly connected, Tunnel0
C   172.16.1.0 is directly connected, FastEthernet0/0
O   172.16.2.0 [110/20] via 172.16.1.2, 00:20:15, FastEthernet0/0
O   192.168.4.0/24 [110/84] via 172.16.1.2, 00:20:15, FastEthernet0/0
192.168.20.0/32 is subnetted, 3 subnets
O   192.168.20.1 [110/65] via 10.1.1.1, 00:20:15, Serial2/0
O   192.168.20.2 [110/65] via 10.1.2.1, 00:20:45, Serial2/1
O   192.168.20.254 [110/65] via 10.1.1.1, 00:20:15, Serial2/0
S   192.168.5.0/24 [110/0] via 10.5.1.1
```

NAT/PAT:

- Overload on Serial2/0
- Inside: FastEthernet0/0
- Outside: Serial2/0

```
LHR-R1#show ip nat statistics
Total active translations: 1 (0 static, 1 dynamic; 1 extended)
Outside interfaces:
  Serial2/0
Inside interfaces:
  FastEthernet0/0
Hits: 161 Misses: 0
CEF Translated packets: 156, CEF Punted packets: 0
Expired translations: 0
Dynamic mappings:
-- Inside Source
[Id: 1] access-list 1 interface Serial2/0 reconfout 1
Appl doors: 0
Normal doors: 0
Queued packets: 0
LHR-R1#
```

5. Verification Results

OSPF Neighbors — HQ-R1:

```
HQ-R1#
HQ-R1#
HQ-R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address        Interface
192.168.20.1    0    INIT/-         00:00:37    172.16.12.2    Tunnel0
192.168.10.2    1    FULL/DR        00:00:37    172.16.1.2     FastEthernet0/0
192.168.60.2    0    FULL/-         00:00:39    10.5.2.1       Serial2/3
192.168.60.1    0    FULL/-         00:00:36    10.5.1.1       Serial2/2
192.168.20.2    0    FULL/-         00:00:38    10.1.2.1       Serial2/1
192.168.20.1    0    FULL/-         00:00:30    10.1.1.1       Serial2/0
HQ-R1#
```

solarwinds | Solar-PuTTY free tool

© 2019-2024 SolarWinds Worldwide, LLC. All rights reserved.

GRE Tunnel:

```
HQ-R1#show interface tunnel 0
Tunnel0 is up, line protocol is up
Hardware is Tunnel
Internet address is 172.16.12.1/30
MTU 1514 bytes, BW 9 Kbit, DLY 5000000 usec,
reliability 255/255, txload 1/255, rxload 1/255
Encapsulation TUNNEL, loopback not set
Keepalive not set
Tunnel source 10.1.1.2 (Serial2/0), destination 10.1.1.1
Tunnel protocol/transport GRE/IP
Key disabled, sequencing disabled
Checksumming of packets disabled
Tunnel TTL 255
Fast tunneling enabled
Tunnel transmit bandwidth 8000 (kbps)
Tunnel receive bandwidth 8000 (kbps)
Last input 00:00:06, output 00:00:05, output hang never
Last clearing of "show interface" counters never
Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
Queueing strategy: fifo
Output queue: 0/0 (size/max)
5 minute input rate 0 bits/sec, 0 packets/sec
5 minute output rate 0 bits/sec, 0 packets/sec
112 packets input, 12104 bytes, 0 no buffer
Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
181 packets output, 19760 bytes, 0 underruns
0 output errors, 0 collisions, 0 interface resets
0 output buffer failures, 0 output buffers swapped out
HQ-R1#
```

Ping Test:

```
HQ-R1#
HQ-R1#ping 192.168.1.1 source loopback0

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.10.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/36/80 ms
HQ-R1#
```

solarwinds | Solar-PuTTY free tool

© 2019-2024 SolarWinds Worldwide, LLC. All rights reserved.

6. Troubleshooting

Issue	Cause	Solution
L2 port IP error	NM-16ESW switch module	Replaced with NM-4T Serial card
NAT memory error	NBAR enabled, low RAM	Disabled NBAR, increased RAM to 256MB
Router not booting	RAM set to 1024MB	Reduced back to 256MB
CDP no neighbors	Routers isolated	Rewired all connections properly